Report on Practical 3

FTP Server

1. Aims

- Learn to set up and configure an FTP server using Cisco Packet Tracer and perform file transfer operations between PCs and the FTP server.

2. Objectives

- Configure network devices like routers, PCs, and an FTP server with their relevant IP address.
- Set up an FTP server with a specific username and password.
- Understand how to transfer files (upload and download) between PCs and the FTP server.
- Checking if the files are present in the FTP server and respective PCs.

3. Background

- FTP - File Transfer Protocol. It is a standard network protocol used for the transfer of files from one host to another over a TCP-based network, such as the Internet.
- FTP can be very useful when transferring/sharing large files like multimedia content, software updates, or scientific data over email is impractical.
- Three primary categories of FTPs:

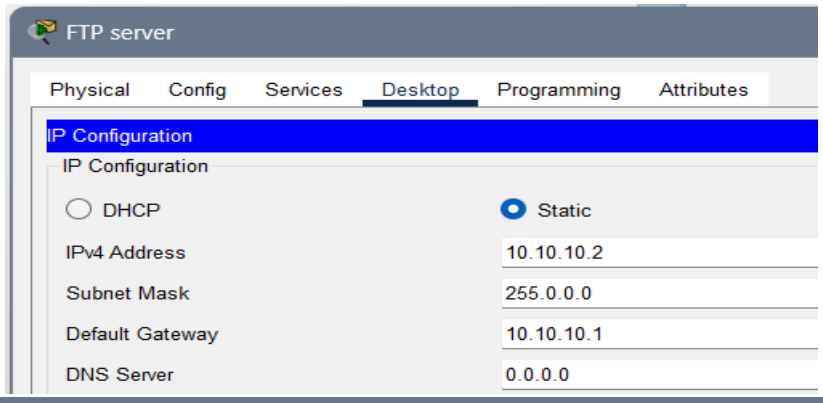
  FTP Plain (normal FTP without encryption)

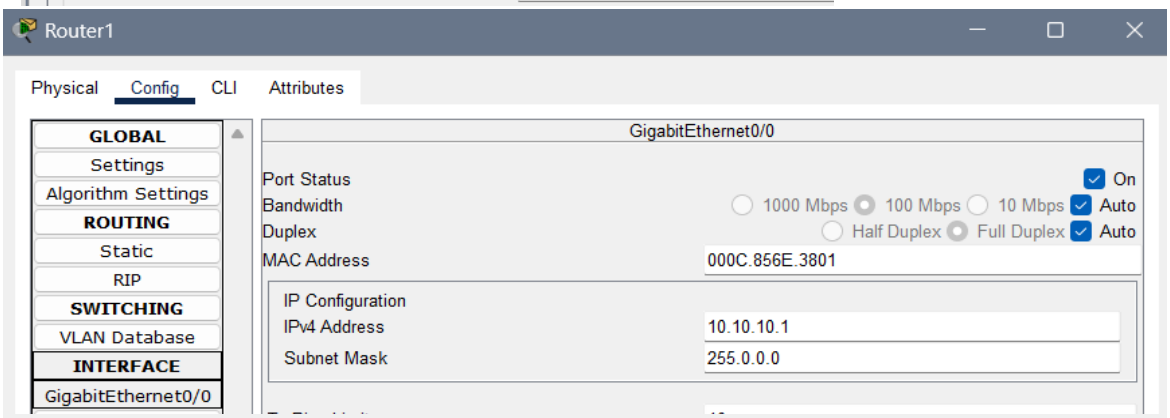
  FTPS (uses SSL encryption)

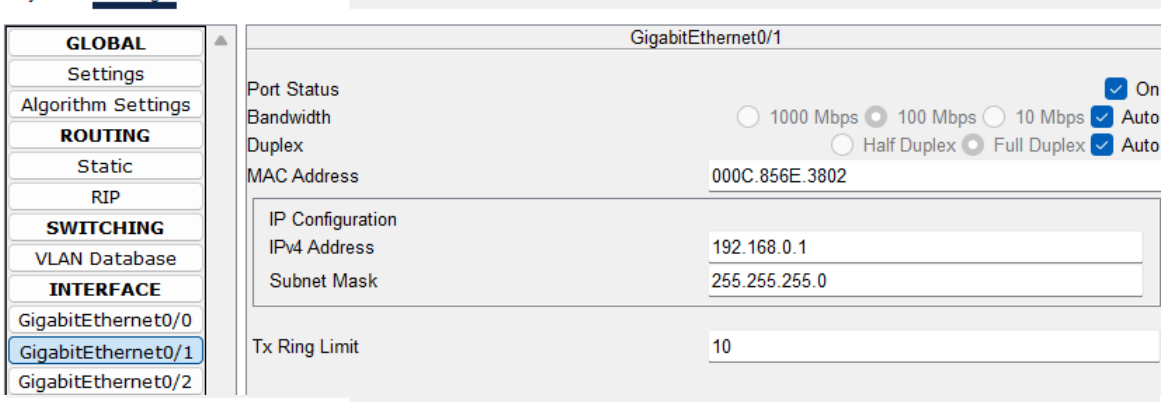
  FTPES (TLS/SSL-encrypted transmission)

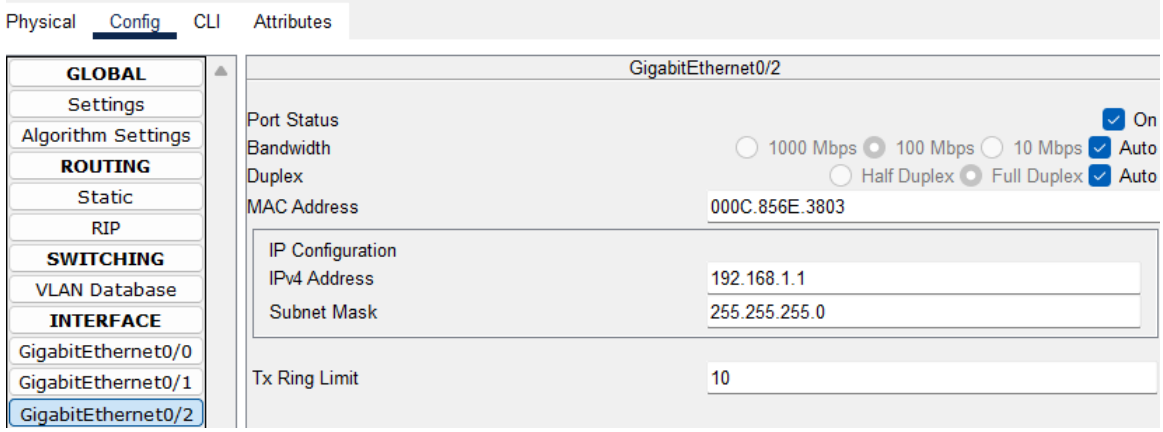
4. Procedure

1. Configure Router1, PC0, PC1, PC2, PC3 and FTP server using relevant IP addresses of your choice.

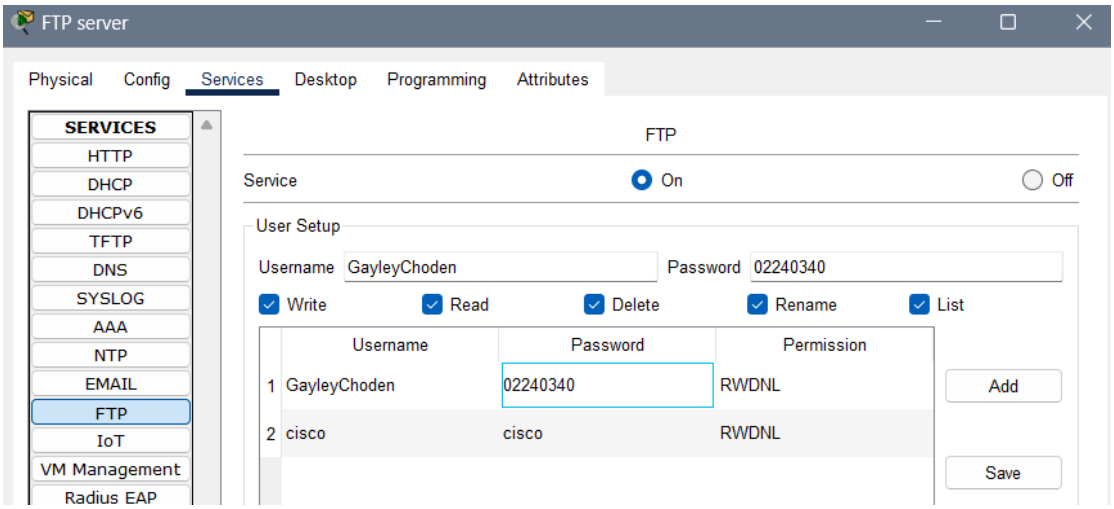

FTP server
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address
10.10.10.2
Subnet Mask
255.0.0.0
Default Gateway
10.10.10.1
DNS Server
0.0.0.0


Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
000C.856E.3801
IP Configuration
IPv4 Address
10.10.10.1
Subnet Mask
255.0.0.0


Physical
Config
CLI
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GigabitEthernet0/1
GigabitEthernet0/2
GigabitEthernet0/1
Port Status
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Bandwidth
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100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
000C.856E.3802
IP Configuration
IPv4 Address
192.168.0.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

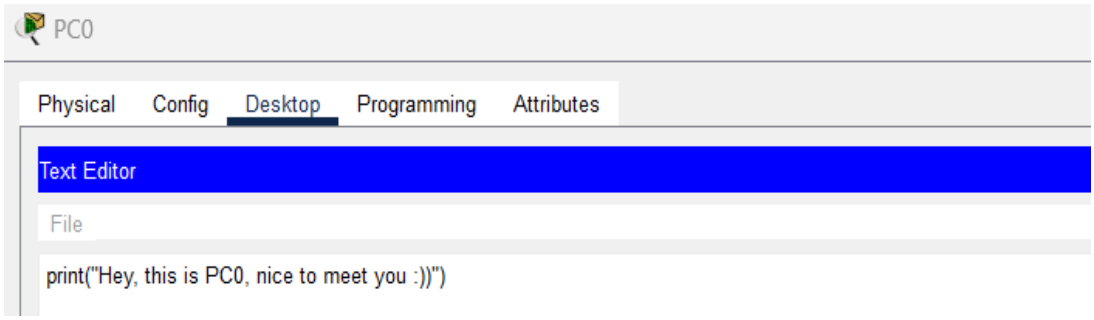

Physical
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Full Duplex
Auto
MAC Address
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IP Configuration
IPv4 Address
192.168.1.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

2. Set up an FTP server

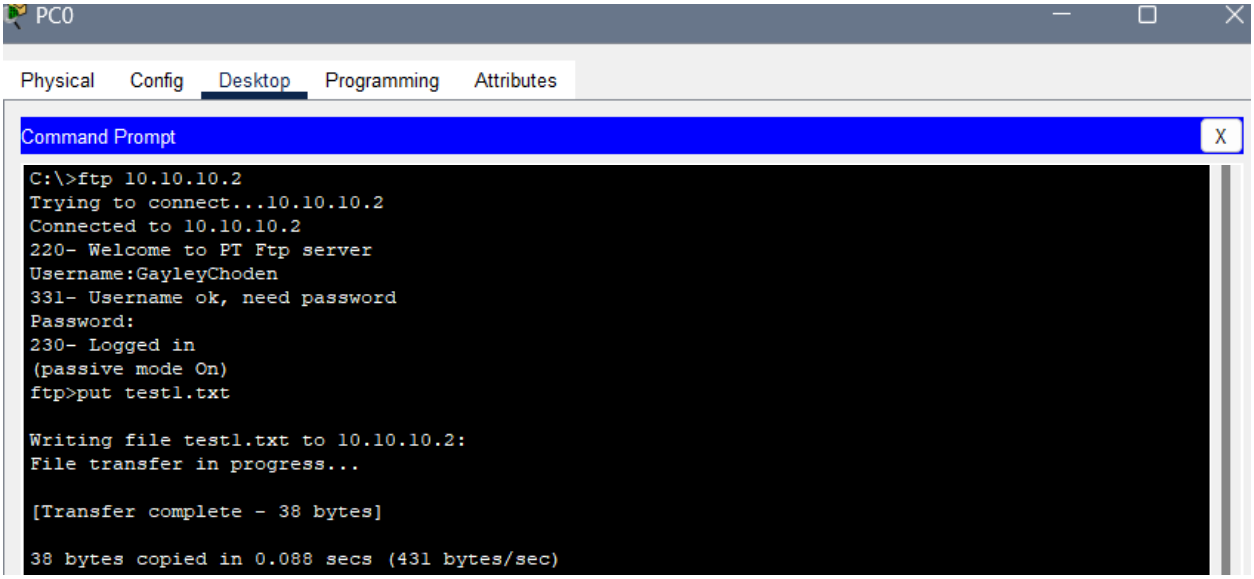


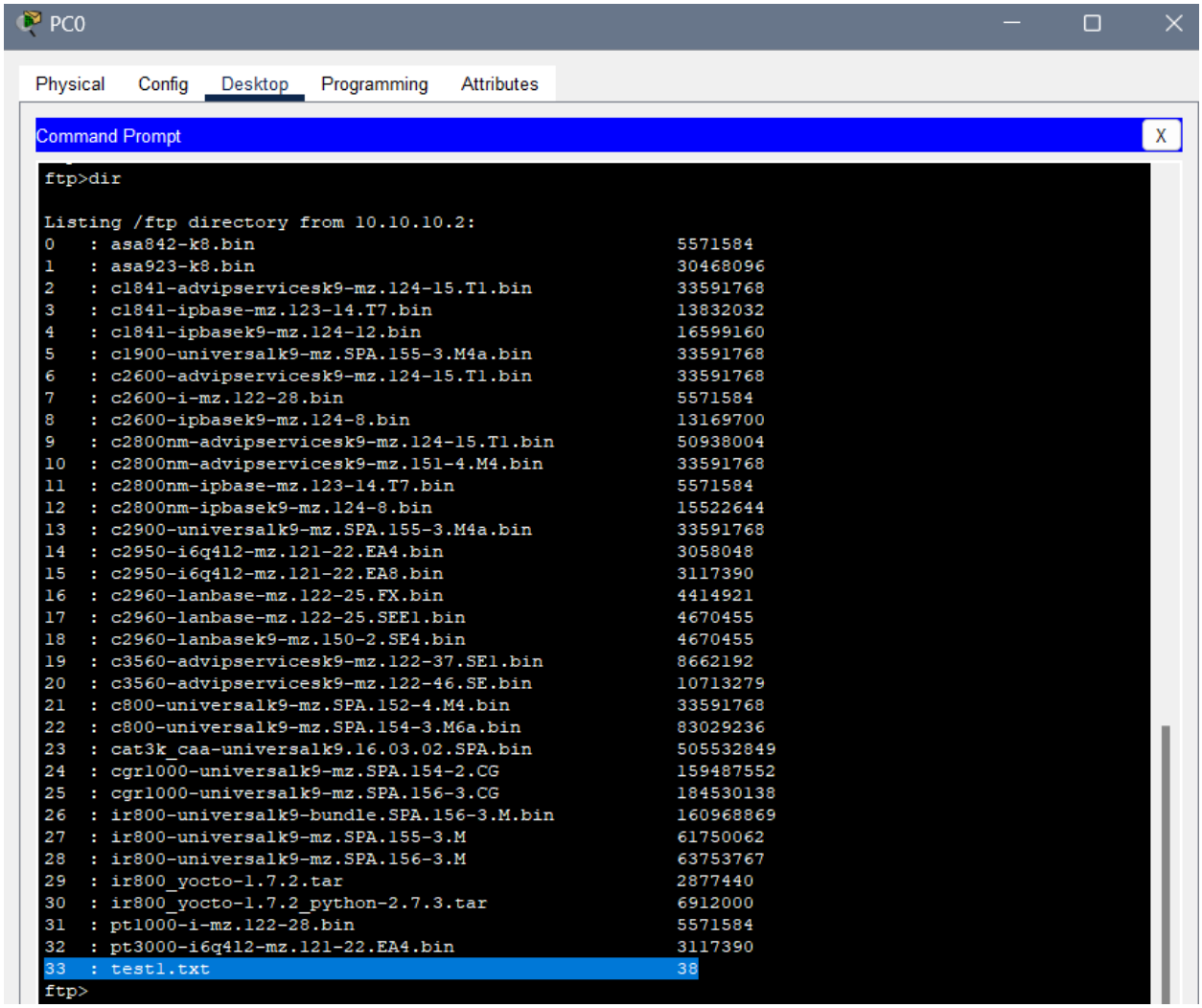
3. On PC0 create and save a **test1.txt** file with the following content:

print("Hey, this is PC0, nice to meet you:))")

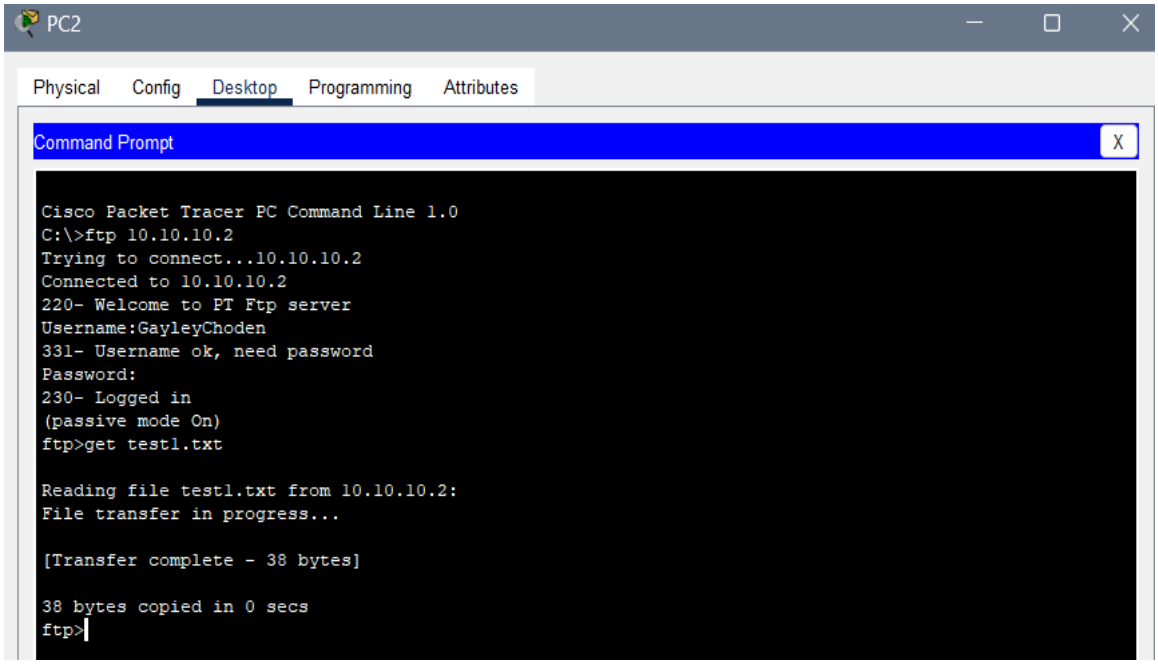


4. Log-in to the FTP server and **PUT** your file into the FTP server. Check if the file is in the FTP server using the **dir** command on the CLI.

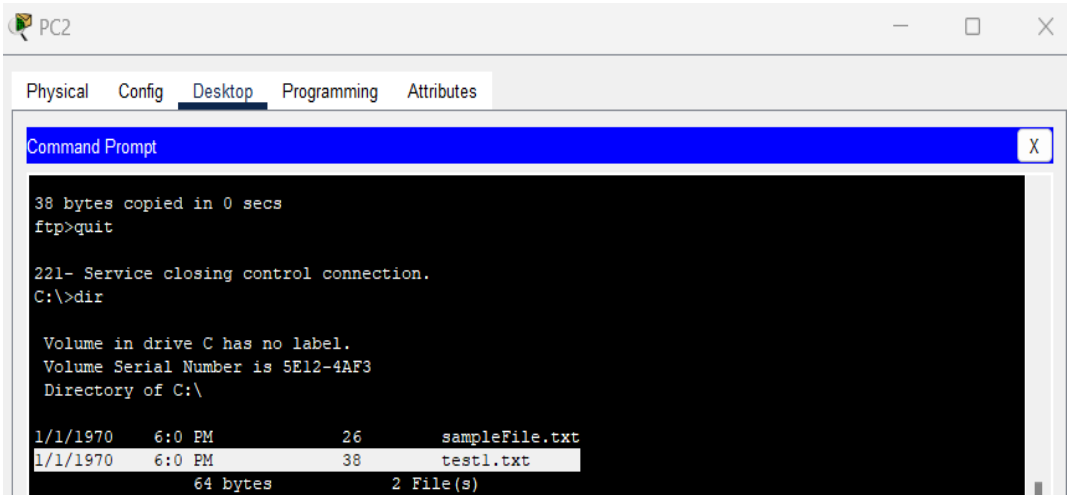




5. Log-in to the FTP server from **PC2** and **GET** the **test1.txt** file.

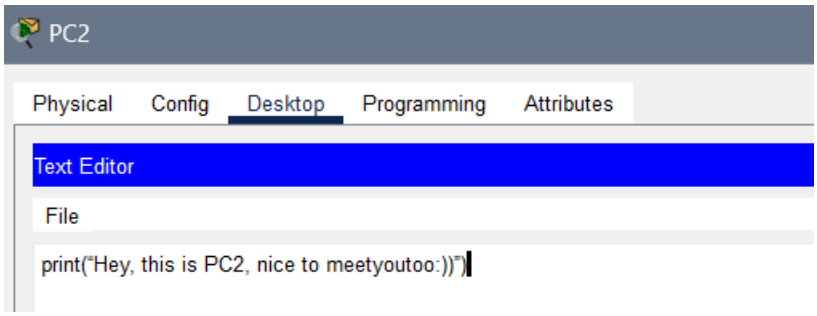


6. Exit the FTP session on **PC2** using the **quit** command and check if the **text1.txt** file is in **PC2** using the **dir** command.

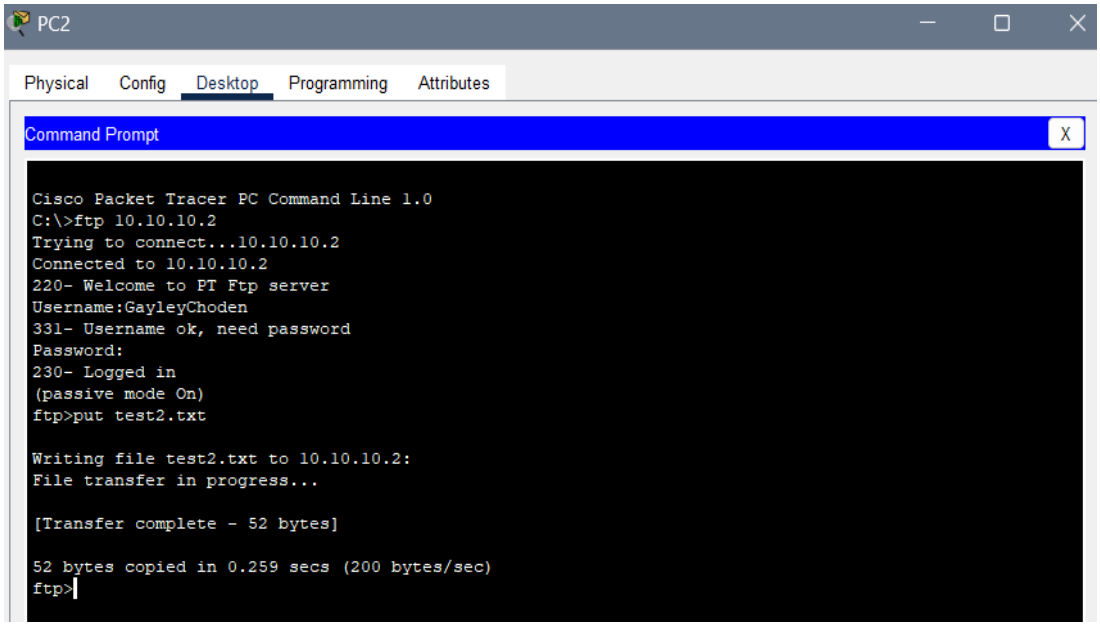


7. Now, create and save a **test2.txt** file on **PC2** with the following content:

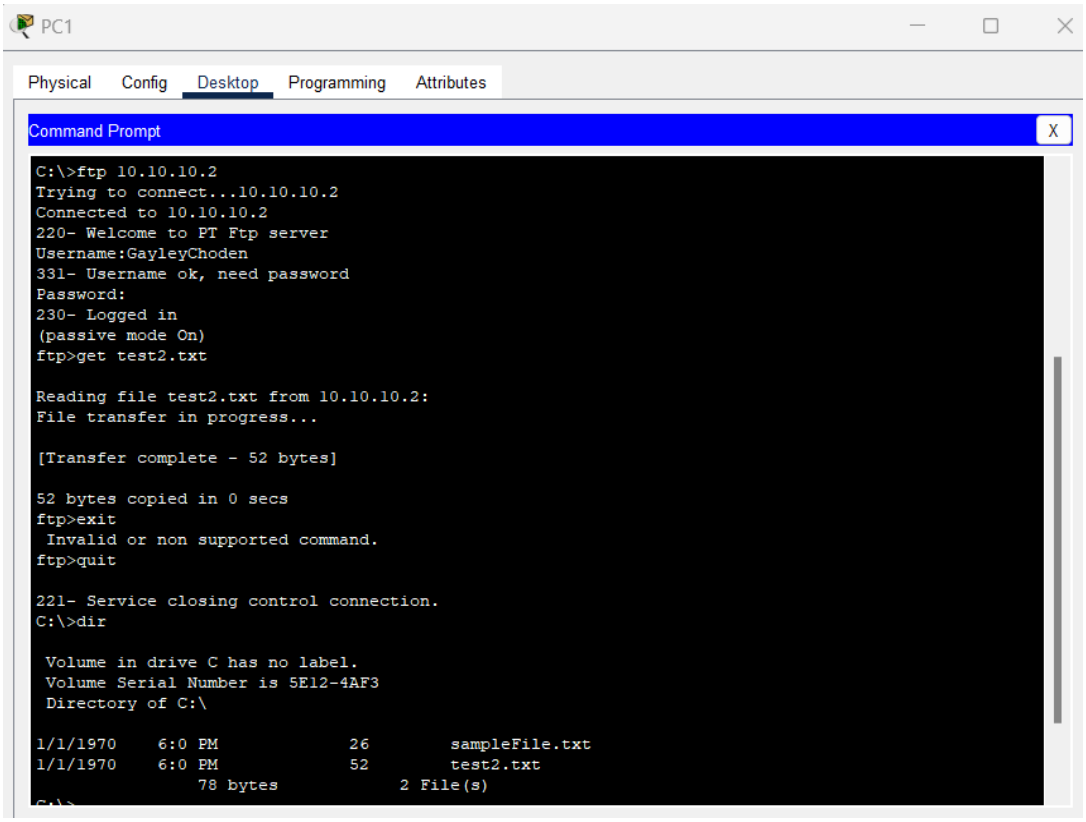
print ("Hey, this is PC2, nice to meet you too:))")



8. **PUT** the file into the FTP server.



9. Login to the FTP server from **PC1** and **GET** the **test2.txt** file from the server. Quit the FTP session and check if the **text2.txt** file is in **PC1** using the **dir** command.



5. Conclusion

The practical work demonstrated the setup and operation of an FTP server in Cisco Packet Tracer. File transfers were conducted using FTP commands like **PUT** and **GET** to upload and transfer file. And we used commands such as **dir** to verify if the file is present in the respective PC and FTP, and **quit** ensured efficient session management. Doing this practical helps us understand the importance of proper network configuration and how FTP can efficiently enable secure and dependable file transfers in a networked environment.

6. References

- YouTube video: FTP Server Using CISCO Packet Tracer

  FTP Server Using CISCO Packet Tracer || CCNA videos easy learning tutorials - YouTube